

# Nexus

## Multi-Specialist AI Research Chatbot

Framework: OpenAI Agents SDK + Streamlit | Version: 1.0 | Date: June 2026  
Repository: bibhu2020/agenticaai (Hugging Face Space)

### 1. Overview

Nexus is a conversational AI assistant that fans out user queries to three specialist sub-agents (Finance, News, Web Research) in parallel, then synthesises their reports into a single, structured response. It runs as a Streamlit web application with persistent SQLite-backed conversation memory and optional Langfuse tracing.

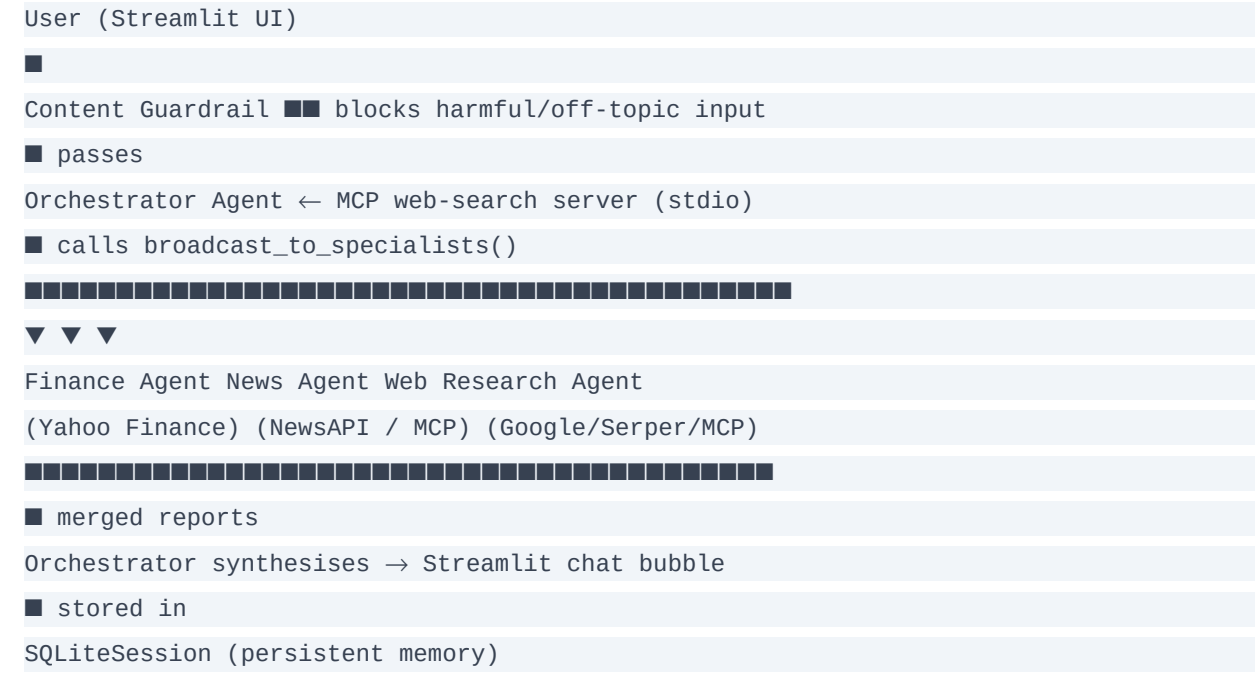
### 2. System Design

#### 2.1 Components

| Component                   | Technology                  | Purpose                                                       |
|-----------------------------|-----------------------------|---------------------------------------------------------------|
| app.py                      | Streamlit                   | UI — chat interface, session state, prompt library            |
| orchestrator.py             | OpenAI Agents SDK Agent     | Central intelligence; owns broadcast_to_specialists tool      |
| specialists.py              | OpenAI Agents SDK Agents x3 | Finance, News, Web specialist agents                          |
| content_guardrail.py        | InputGuardrail              | Blocks off-topic or harmful prompts before they reach the LLM |
| model_factory.py / llm.py   | openai.AsyncOpenAI          | LLM client factory — supports openai, google, groq, ollama    |
| MCP server (mcp-web-search) | MCPServerStdio              | Web-search capability exposed via Model Context Protocol      |

| Component        | Technology                     | Purpose                                                        |
|------------------|--------------------------------|----------------------------------------------------------------|
| SQLiteSession    | OpenAI Agents<br>SQLiteSession | Persistent cross-session conversation history                  |
| Langfuse tracing | tracing.py                     | Optional observability; disabled when LANGFUSE keys are absent |

2.2 Architecture Diagram



3. Agentic Flow

Each user turn follows this deterministic sequence:

| Step | Actor                           | Action                                                             |
|------|---------------------------------|--------------------------------------------------------------------|
| 1    | Content Guardrail               | Evaluate input; tripwire if off-topic or unsafe                    |
| 2    | Orchestrator                    | Parse intent; decide which specialist(s) are needed                |
| 3    | broadcast_to_specialists (tool) | Fan out query to selected agents concurrently (asyncio.gather)     |
| 4    | Finance / News / Web Agents     | Execute tools (market data, news fetch, web search); return report |
| 5    | Orchestrator                    | Synthesise all reports into structured markdown response           |
| 6    | Streamlit                       | Render response; persist turn in SQLiteSession                     |

4. Tools & Integrations

- **Finance Agent:** Yahoo Finance (yfinance) for real-time prices, earnings, analyst ratings

- **News Agent:** NewsAPI for headlines and recent events by topic or category
- **Web Research Agent:** Google Search (Serper) / MCP web-search server for general knowledge
- **Content Guardrail:** OpenAI Agents InputGuardrail running a lightweight classifier agent
- **MCP Server:** External web-search tool server connected via stdio (MCPServerStdio)
- **Langfuse:** Distributed tracing for debugging and monitoring agent calls

## 5. Configuration

| Env Var                             | Default | Required for         |
|-------------------------------------|---------|----------------------|
| OPENAI_API_KEY                      | —       | Default LLM provider |
| GOOGLE_API_KEY                      | —       | Gemini model option  |
| GROQ_API_KEY                        | —       | Groq model option    |
| SERPER_API_KEY                      | —       | Google web search    |
| NEWS_API_KEY                        | —       | News agent           |
| LANGFUSE_SECRET_KEY /<br>PUBLIC_KEY | —       | Optional tracing     |